

Analysis of an iron ore

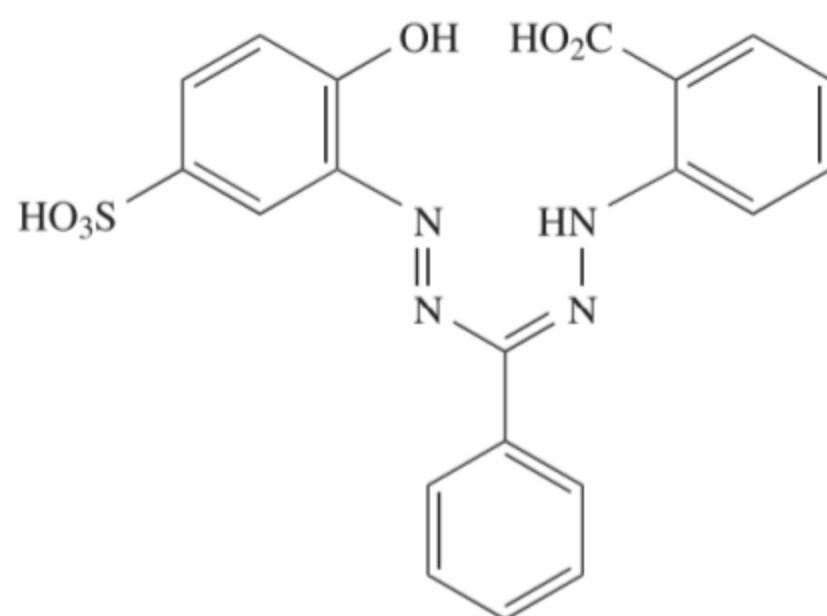
The iron content of hematite, an iron (III)-oxide containing ore, can be determined by an oxidimetric titration. In such an analysis 4.00 g of powdered hematite were treated with concentrated hydrochloride acid. The insoluble parts were filtered off, the remainder was washed carefully, and all the iron ions in the filtrate were reduced to iron (II)-ions. The solution ($\text{pH} = 0$) was titrated with a dichromate titrant ($c = 0.100 \text{ mol/L}$). The titration volume then was 33.74 mL.

- 1.1. Write a balanced reaction equation for the reaction of the iron oxide with hydrochloric acid.
- 1.2. Choose from the following list and argue which of the reducing agents are suitable to reduce to Fe (II): bromide, nitrite, sulphite, zinc (Jones-reductor), tin (II)-solution.
- 1.3. Write a balanced reaction equation for the redox titration.
- 1.4. Calculate the mass percentage on iron (III)-oxide in the sample.
- 1.5. Calculate the equilibrium constant for the titration reaction.
- 1.6. Calculate the potential of the equivalent point of the titration reaction at the given conditions.
- 1.7. Which of the given redox indicators are suitable to indicate in this analytical determination? Give reasons for your assumption. Iron phenanthroline (pale blue/red – $E^\circ = 1.14\text{V}$), ferroine (blue/orange red – $E^\circ = 1.06\text{V}$), indigocarmin (blue/yellow – $E^\circ = 0.29\text{V}$), methylenblue (blue/colorless – $E^\circ = 0.53\text{V}$), nilblue sulphate (violet/colorless – $E^\circ = 0.41\text{V}$)
- 1.8. Is it possible that chloride ions will be oxidized in this titration? Give reasons for your assumption.

Another possibility of the quantitative determination of iron (II)-solutions would be the titration with potassium permanganate solution. In order to avoid the generation of chlorine in this method, 10.0 mL of the so-called Reinhardt-Zimmermann-reagent are added. This is a solution of manganese (II)-sulphate in a mixture of sulfuric and of phosphoric acid with relatively high concentrations.

- 1.9. Calculate the minimum pH-value at which chlorine will be generated in the titration with permanganate (all the other ions which influence the potential have the activity $a = 1$).
- 1.10. In which direction will the standard potentials of higher oxidized manganese species change, if manganese (II)-sulphate is added.
- 1.11. What is the effect of adding phosphoric acid?

2. The compound shown below is a formazan dye usually referred to as 'zincon'. It is used to detect Zn^{2+} and Cu^{2+} ions:



- 2.1 Suggest how the ligand binds to Zn^{2+} and Cu^{2+} , and comment on the role of pH in determining the overall charge of the complex.
- 2.2 Why does the ligand include a SO_3H substituent?
- 2.3 Zincon itself absorbs at 463 nm. Suggest how the absorption arises.
- 2.4 When zincon binds Cu^{2+} , the absorption at 463 nm is replaced by one at 600 nm. Why does this make zincon an easy method of detection for Cu^{2+} ions?
- 2.5 The copper(II) complex of zincon can be used as a sensor for $[\text{CN}]^-$ ions in aqueous solution. Addition of $[\text{CN}]^-$ results in the disappearance of the absorption at 600 nm and reappearance of the absorption at 463 nm. Outline the chemical changes occurring in solution.

3. Heavy Metal Detoxification

2,3-dimercaptosuccinic acid (DMSA) is a medication for metal poisoning, such as lead (Pb), mercury (Hg), and arsenic (As), with the formula $\text{HO}_2\text{CCH}(\text{SH})\text{CH}(\text{SH})\text{CO}_2\text{H}$. Its mechanism of action is based on chelating by which the drug, called a chelating agent, strongly binds the metal ion, forming a water soluble 1:1 complex that is then excreted via the urinary system.

Dicarboxylicacetylene A is reacted with thioacetic acid to give compound B to yield the product DMSA.



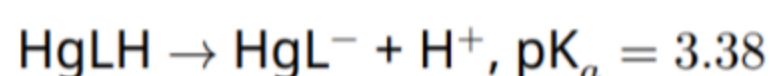
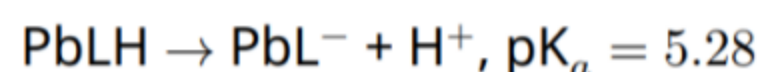
- 3.1 Write this reaction sequence with structures of all compounds and the reagent X.
- 3.2 DMSA has several stereoisomers. Draw all of them, give them a number, and state the absolute

configuration of each chiral center. Are there sets of enantiomers within the possible isomers? If yes, indicate which of them.

3.3 The active substance in the medication is only one isomer. This isomer reveals a rotation angle of zero under the polarimeter. Give the isomer that represents the active substance in the pharmaceutical.

3.4 DMSA is a tetraprotic acid. Its acid dissociation constants are: $pK_{a1} = 2.71$; $pK_{a2} = 3.48$; $pK_{a,SH1} = 9.65$; $pK_{a1,SH2} = 12.05$. Calculate the pH of a 0.1 mol/L solution of uncoordinated DMSA (The autoprotolysis of water may be neglected and the deprotonation steps can be considered decoupled).

The acid dissociation constants of the uncoordinated groups in the Pb^{2+} and Hg^{2+} chelates of mono-O-methyl DMSA are given:



The IR spectrum of $HgLH$ shows the presence of a protonated carboxylic group, that of $PbLH$ the presence of a carboxylate group.

3.5 Draw the structure of the Pb and of the Hg complex.

3.6 Explain the different pK_a values.

A person was diagnosed with Pb-poisoning when a blood test revealed a blood lead level (BLL) of 85 $\mu\text{g/dL}$. The patient was treated with intravenously administered DMSA and the given dose corresponded to 5 mg/kg. Assume an average body weight of 75 kg and blood volume of 5 L.

3.7 Assuming that the equilibrium concentration of DMSA is 99.5% of the administered dose, determine the free Pb concentration after equilibrium has been reached ($t = 0$).

The clearance of the $Pb(\text{DMSA})$ complex was then monitored by the concentration change in free Pb over a period of 6 hours. The results are shown in the table below:

t / min	0	60	120	180	240	300	360
c / $\mu\text{mol/L}$	c from task 7	1.66	1.34	1.08	0.87	0.70	0.56

(The values are simplified)

3.8 Calculate the apparent $t_{1/2}$ and k for the clearance of free Pb from the patients' blood.

3.9 Calculate how long it will take until the limiting value of 90 $\mu\text{g/L}$ free Pb is reached.

To test the interaction of the DMSA complex with Cd^{2+} , another toxic metal ion that binds to this ligand, the following experiment was carried out: To 0.1 mmol DMSA-Pb in 1 L of neutral water, 0.05 mmol of a Cd(II) salt was added. Assume that the volume did not change and no acid/base reaction occurred.

3.10 Write the reaction equation and the expression for the resulting equilibrium.

3.11 Calculate the equilibrium constant. The stability constant for DMSA-Pb is $K_1 = 1017.6 \text{ M}^{-1}$ and for DMSA-Cd $K_2 = 1016.5 \text{ M}^{-1}$.

3.12 Calculate the equilibrium concentrations of all four species.

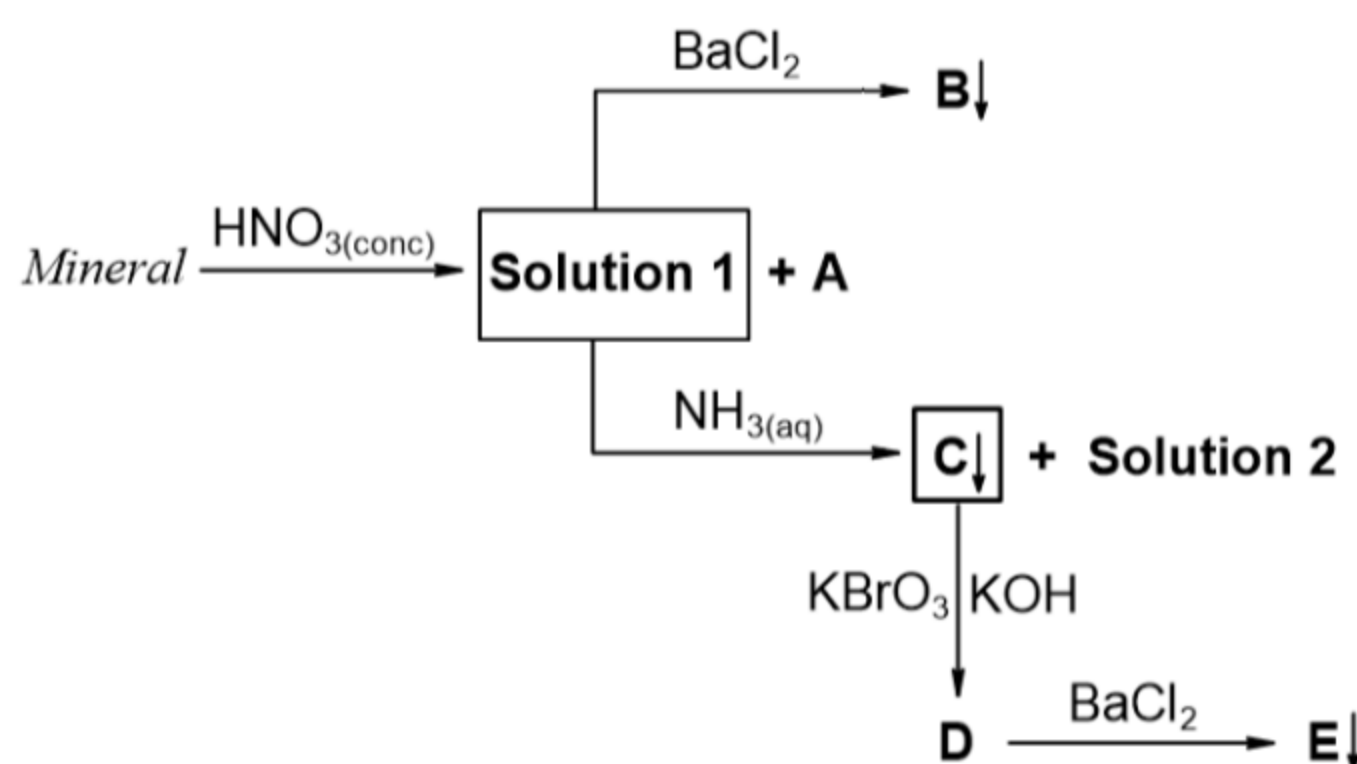
4. Avicenna

The philosopher Avicenna took a vial with a strange brownish-bronze mineral inside. Avicenna performed many reactions with this mineral and created a scheme to help to clearly identify the composition of the mineral.

Avicenna dissolved 1.00 g of the mineral in an excess of concentrated nitric acid and observed a release of a brown gas **A** with a volume of 1.65 L under standard conditions (0 °C, 1 atm).

The solution formed (**Solution 1**) was divided into two equal parts for further analysis. To the first half of **Solution 1**, he added an excess of BaCl_2 , which led to 0.929 g of white precipitate **B**.

He added excess aqueous ammonia solution to the second half of **Solution 1**, causing the formation of brown precipitate **C** and deep-blue **Solution 2**. **C** was filtered off, dissolved in an alkaline solution of KBrO_3 , giving a purple solution of compound **D**. **D** reacted with BaCl_2 to form 0.256 g of the red precipitate **E**.



1. **Determine** the possible elemental composition of the mineral.
2. **Determine** the molecular formula of the mineral and compounds **A–E**. **Show** your calculations where necessary.
3. **Write** all the reaction equations described in the scheme.